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Professors Daniel Ballin and Robert D. Macredie

Editors-in-Chief

Virtual Reality

Dear Professors Ballin and Macredie,

Please consider our Original Article, "**Viscereality: Bioresponsive Virtual Reality for Box Breathing and Altered-State Experience**," for publication in *Virtual Reality*.

Breath-based VR interfaces typically render respiration through discrete objects or physiological proxies. We developed Viscereality to examine whether respiration can instead organise the surrounding immersive space through a flexible particle field, making breathing continuously perceptible as an environmental transformation.

We evaluated the system in a preregistered within-subjects study in which 39 participants completed symmetric VR, asymmetric VR, and dark-screen control sessions during slow-paced box breathing. Relative to the control, the pooled VR conditions improved breathing synchronisation, increased the perceived spatial extension of the self, and produced altered-state experiences characterised primarily by perceptual effects. We found no reliable experiential difference between the particle-symmetry conditions. The results therefore distinguish the effects of the overall immersive paradigm from the unresolved contribution of the specific aesthetic manipulation.

The study connects breath-based interaction, biofeedback, embodiment, and immersive spatial design. It shows how a weakly representational particle medium can function as an affective interface and provides a controlled evaluation of its effects on breathing performance and subjective experience. This contribution directly fits the journal's multidisciplinary focus on the development and evaluation of VR systems, techniques, and applications.

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission to *Virtual Reality*.

The authors declare no competing interests. Versioned data, analysis code, and reproducibility materials accompany the manuscript.

Thank you for considering our manuscript.

Sincerely,

George Fejer

Corresponding author, on behalf of all authors